

Quiz 0

Both **regression** and **classification** tasks are a form of predictive modeling approach to characterize the relationship between observational input data and a desired response (or target). In other words, they are both **supervised learning** tasks. In classification, the target is categorical, whereas in regression, the target is continuous.

Examples of classification tasks:

- Handwritten character classification
- Speech emotion classification

Examples of regression tasks:

- Car detection in an image
- Mortgage rate prediction

```
In [1]: import numpy as np
import matplotlib.pyplot as plt
%matplotlib inline
plt.style.use('bmh')
```

```
In [2]: X = np.load('dataset.npy')

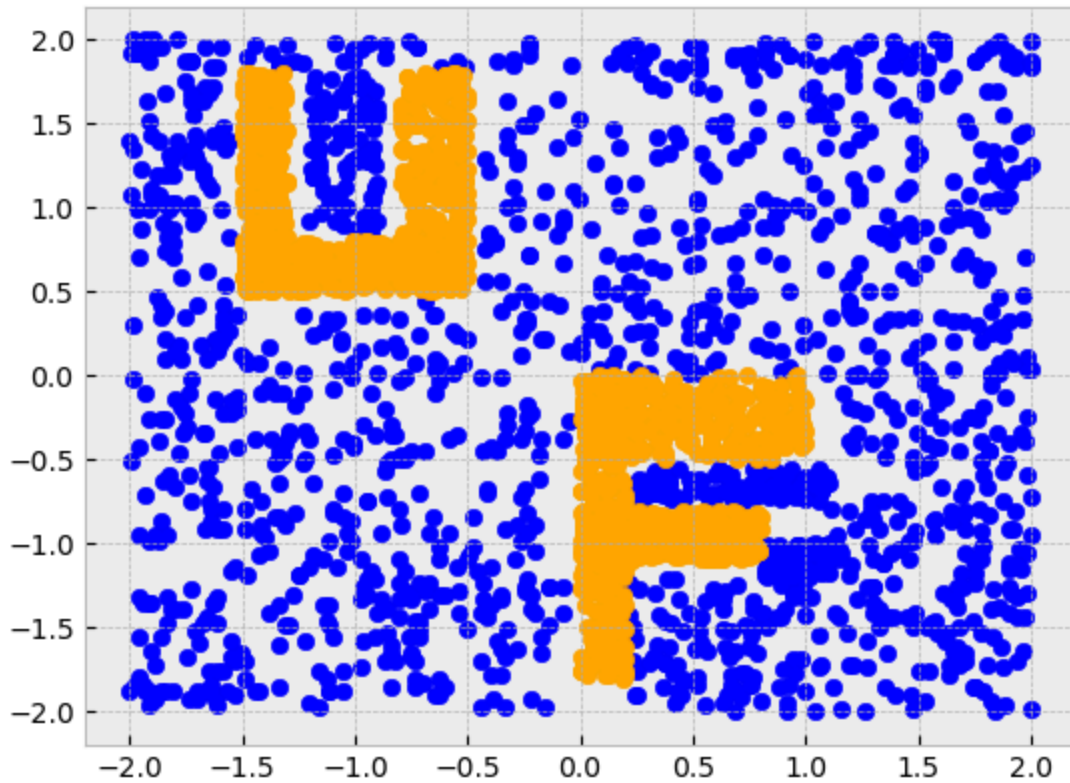
X.shape
```

```
Out[2]: (3000, 3)
```

The dataset is a matrix of size 3000×3.

```
In [3]: plt.scatter(X[X[:,2]==0,0],X[X[:,2]==0,1], c='blue')
plt.scatter(X[X[:,2]==1,0],X[X[:,2]==1,1], c='orange');

# several color-coding will be accepted for full credit
```



The dataset represents a UF mask.

In [4]: `import scipy.stats as stats`

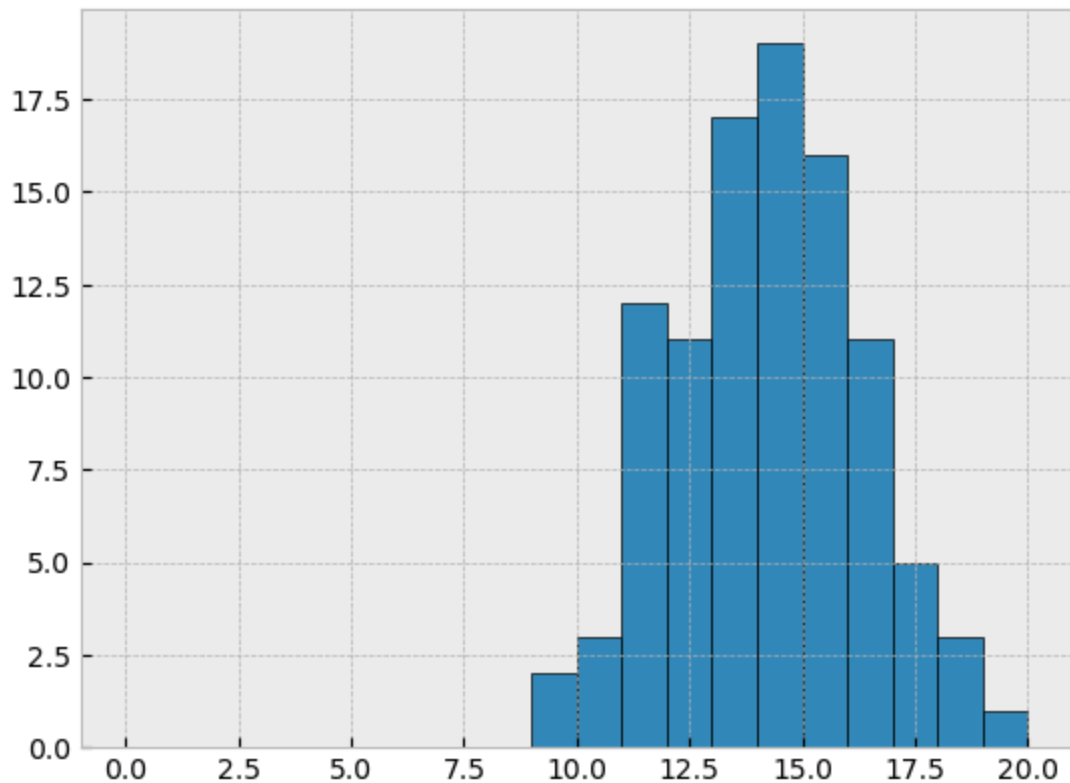
```
In [5]: # Binomial RV with parameters N=20, p=0.7
X = stats.binom(20,0.7)

# Draw 100 samples at random from this RV
sample = X.rvs(size=100)

# Plot the histogram
plt.hist(sample, bins=range(0,21), edgecolor='k');

# Under this RV, probability that observed sample is >=15 is
print('Under this RV, probability that observed sample is >15 is', X.sf(15))
```

Under this RV, probability that observed sample is >15 is 0.23750777887760147



Course Policies

Please read the course [syllabus](#) for a full description of course policies.

Prerequisites Mini-Test

1. Answer the following questions:

- A. Consider the function, $f(x) = \frac{1}{2}(g(x))^4$, where $g(x) = 2x + 1$.
 Use the chain rule to compute the derivative $\frac{\partial f}{\partial x}$.
 Show your work (you may include a picture or scan of your handwritten solution). Points will be deducted if you use back substitution.
- B. Consider the function, $f(x_1, x_2) = x_1^2 x_2 + x_1 x_2^3$ in $\mathbb{R}^2$. Compute the gradient of $f(\mathbf{x})$, with $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \in \mathbb{R}^2$.

Question 1.A

Applying the chain rule, we have:

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial g} \frac{\partial g}{\partial x}$$

where

$$\begin{aligned} \frac{\partial f}{\partial g} &= \frac{1}{2} \times 4 \times (g(x))^3 = \\ 2(g(x))^3 \end{aligned} \quad \frac{\partial g}{\partial x} = 2$$

Putting it together:

$$\frac{\partial f}{\partial x} = 4(g(x))^3$$

Question 1.B

The gradient of $f(\mathbf{x})$ is:
$$\begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \end{bmatrix} = \begin{bmatrix} 2x_1x_2 + x_2^3 \\ x_1^2 + 3x_1x_2^2 \end{bmatrix}$$

2. What is the rank of a matrix? What is the rank for the following two matrices:

$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ and

$B = \begin{bmatrix} 1 & 2 & 1 \\ -2 & -3 & 1 \\ 3 & 5 & 0 \end{bmatrix}$.

Let $A \in \mathbb{R}^{m \times n}$ where $n \neq m$. The rank of the matrix A is defined as $\text{rk}(A)$ and corresponds to the largest number of linearly independent rows or columns, that is, $\text{rk}(A) \leq \min(m, n)$.

When $\text{rk}(A) = \min(m, n)$, the matrix A is said to be **full rank**. We can write the matrix A in reduced row echelon form (RREF) to count how many rows have a leading 1.

Alternatively, if the matrix is square ($m=n$), we can compute the determinant, $\det(A)$. If $\det(A) \neq 0$, then the square matrix A is full rank.

For the examples provided, we find that the third row (or column) of matrix $A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ is a linear combination of the first 2 rows (or columns). Thus the rank is less than 3. Since A is already written in RREF, we find that it contains 2 linearly independent columns. Thus $\text{rk}(A) = 2$.

We can also write the matrix B in RREF to check how many independent columns/rows it contains. To do this, I will use `sympy`, but multiple other approaches will be acceptable.

```
In [6]: import sympy
from sympy import Matrix
import numpy as np

B = np.array([[1, 2, 1], [-2, -3, 1], [3, 5, 0]])

Matrix(B).rref()
```

```
Out[6]: (Matrix([
[1, 0, -5],
[0, 1, 3],
[0, 0, 0]]),
(0, 1))
```

Similarly, we find that $\text{rk}(B)=2$. Alternatively, not that B is a square matrix so we can compute the determinant to check if it's different than 0.

```
In [7]: np.linalg.det(B)
```

```
Out[7]: 8.881784197001244e-16
```

If a square matrix is not full rank, in addition to a determinant equal to 0, at least one of the eigenvalues is 0. Let's verify:

```
In [8]: L, U = np.linalg.eig(B)
        L, U
```

```
Out[8]: (array([-3.82842712e+00,  2.54905302e-16,  1.82842712e+00]),
         array([[ 0.11070323, -0.84515425,  0.64522571],
                [-0.64522571,  0.50709255, -0.11070323],
                [ 0.75592895, -0.16903085,  0.75592895]]))
```

3. Consider the vectors

$\mathbf{a} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$ and

$\mathbf{b} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$.

A. Find the vector projection of $\mathbf{b}$ onto $\mathbf{a}$, i.e.,

$\text{proj}_{\mathbf{a}} \mathbf{b}$.

B. Is the error of projection orthogonal to vector $\mathbf{a}$? Justify your answer.

Question 3.A

The vector projection of $\mathbf{b}$ onto $\mathbf{a}$ is:

$$\text{proj}_{\mathbf{a}} \mathbf{b} = \frac{\mathbf{b}^T \mathbf{a}}{\|\mathbf{a}\|^2} \mathbf{a}$$

where $\|\mathbf{a}\| = \sqrt{1^2 + 2^2 + 2^2} = 3$, $\|\mathbf{a}\|^2 = 9$,

$\mathbf{a}^T \mathbf{b} = 1 \times 1 + 1 \times 2 + 1 \times 2 = 5$, hence

$\frac{\mathbf{b}^T \mathbf{a}}{\|\mathbf{a}\|^2} = \frac{5}{9}$. Therefore, we have:

$$\text{proj}_{\mathbf{a}} \mathbf{b} = \frac{5}{9} \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} \frac{5}{9} \\ \frac{10}{9} \\ \frac{10}{9} \end{bmatrix}$$

Question 3.B

The error of projection is defined as:

$$\mathbf{e} = \mathbf{b} - \text{proj}_{\mathbf{a}} \mathbf{b} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} \frac{5}{9} \\ \frac{10}{9} \\ \frac{10}{9} \end{bmatrix} =$$

$$\begin{bmatrix} \frac{4}{9} & -\frac{1}{9} & -\frac{1}{9} \end{bmatrix}$$
 \end{align*}

The inner product between the error of projection vector $\mathbf{e}$ and the vector $\mathbf{a}$, $\mathbf{e}^T \mathbf{a} = 0$, thus they are orthogonal.

4. Consider repeatedly rolling a fair 6-sided die, and let E_i be the event that you observe a 6 on roll i . How many rolls of the die would you have to do to have a 90% probability that you would see at least one 6?

$$P\left(\bigcup_{i=1}^n E_i\right) \geq 0.9 \quad \parallel \quad 1 - P\left(\bigcap_{i=1}^n \overline{E_i}\right) \geq 0.9, \text{by DeMorgan's Law} \quad \parallel \quad 1 - \prod_{i=1}^n P(\overline{E_i}) \geq 0.9, \text{by statistical independence (fair die)} \quad \parallel \quad 1 - \prod_{i=1}^n \left(1 - \frac{1}{6}\right) \geq 0.9 \quad \parallel \quad 1 - \left(\frac{5}{6}\right)^n \geq 0.9 \quad \parallel \quad \left(\frac{5}{6}\right)^n \leq 0.1 \quad \parallel \quad n \times \log\left(\frac{5}{6}\right) \leq \log(0.1) \quad \parallel \quad n \geq 12.6 \quad \rightarrow \quad n = 13$$

Check: $1 - \left(\frac{5}{6}\right)^{13} \approx 0.906 \geq 0.9$.

5. Let $\mathbf{A}$ and $\mathbf{B}$ be events with positive probability. Prove that $P(\mathbf{A}|\mathbf{B}) > P(\mathbf{A})$ if and only if $P(\mathbf{B}|\mathbf{A}) > P(\mathbf{B})$.

From Bayes' Theorem, we have that $P(\mathbf{A}|\mathbf{B}) = \frac{P(\mathbf{B}|\mathbf{A})P(\mathbf{A})}{P(\mathbf{B})}$. If $P(\mathbf{B}|\mathbf{A}) > P(\mathbf{B})$, then $\frac{P(\mathbf{B}|\mathbf{A})}{P(\mathbf{B})} > 1$, thus $P(\mathbf{A}|\mathbf{B}) > P(\mathbf{A})$.

Similarlay, from Bayes' Theorem, we also have $P(\mathbf{B}|\mathbf{A}) = \frac{P(\mathbf{A}|\mathbf{B})P(\mathbf{B})}{P(\mathbf{A})}$. If $P(\mathbf{A}|\mathbf{B}) > P(\mathbf{A})$, then $\frac{P(\mathbf{A}|\mathbf{B})}{P(\mathbf{A})} > 1$, thus $P(\mathbf{B}|\mathbf{A}) > P(\mathbf{B})$.

This proves that $P(\mathbf{A}|\mathbf{B}) > P(\mathbf{A})$ if and only if $P(\mathbf{B}|\mathbf{A}) > P(\mathbf{B})$.

6. A computer manufacturer uses chips from three sources. Chips from sources A, B, and C are defective with probabilities 0.001, 0.005, and 0.01, respectively. Let P_A , P_B , and P_C denote the probabilities that a chip is from source A, B, or C, respectively.

Let:

- D be the event that a chip is defective,
- A be the event that the chip is from manufacturer A,
- B be the event that the chip is from manufacturer B, and
- C be the event that the chip is from manufacturer C.

We are given that: $P(D|A) = 0.001$, $P(D|B) = 0.005$ and $P(D|C) = 0.01$.

If the computer manufacturer gets 10% of its chips from source A, 20% of its chips from source B, and 70% of its chips from source C, find the (conditional) probability that it came from each source. Show all your work.

We are given different ratios for chips coming from manufacturers A, B and C, respectively: $P(A) = 0.1$, $P(B) = 0.2$ and $P(C) = 0.7$.

Under these priors, the probability that a randomly chosen chip is defective is:

$$\begin{aligned} P(D) &= P(D|A)P(A) + P(D|B)P(B) + P(D|C)P(C) \\ &= 0.001 \times 0.1 + 0.005 \times 0.2 + 0.01 \times 0.7 \\ &\approx 0.0081 \end{aligned}$$

Furthermore, the probability that the defective chip came from each source is calculated using the Bayes' Theorem:

$$\begin{aligned} P(A|D) &= \frac{P(D|A)P(A)}{P(D)} = \frac{0.001 \times 0.1}{0.0081} \approx 0.0123 \\ P(B|D) &= \frac{P(D|B)P(B)}{P(D)} = \frac{0.005 \times 0.2}{0.0081} \approx 0.123 \\ P(C|D) &= \frac{P(D|C)P(C)}{P(D)} = \frac{0.01 \times 0.7}{0.0081} \approx 0.864 \end{aligned}$$

Check: $P(A|D) + P(B|D) + P(C|D) \approx 1$.

7. Let X be a discrete random variable with the following PMF:

$$p_X(x) = \begin{cases} \frac{x^2}{a}, & \text{if } x = -3, -2, -1, 0, 1, 2, 3 \\ 0, & \text{otherwise} \end{cases}$$

Find the first order moment of X .

The first order moment of X is its expected value, or the mean. The expected value is:

$$\begin{aligned} E[X] &= \sum_{k=-3}^3 x p_X(x) \\ &= -3 \times \frac{9}{28} - 2 \times \frac{4}{28} - 1 \times \frac{1}{28} + 0 + 1 \times \frac{1}{28} + 2 \times \frac{4}{28} + 3 \times \frac{9}{28} \\ &= 0 \end{aligned}$$

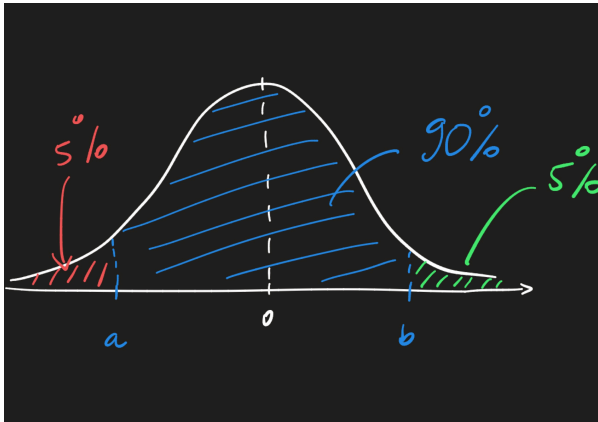
8. Let $\mathbf{X}$ be a continuous random variable modeled by a Gaussian distribution with mean 0 and variance 1 (normal distribution), that is, $\mathbf{X} \sim G(0,1)$. Under this random variable, what values of a and b should you use such that $P(a \leq X \leq b) = 0.9$? You may use the [Q-function table](#) to help you find these values. The Q-function computes the survival function of a normal distribution.

Note that there are multiple answers for a and b . The following answer will require symmetry around the mean. Under this requirement, the solution is unique.

Remember that Gaussian distribution is symmetric with respect to the mean ($\mu=0$). If $P(a \leq X \leq b) = 0.9$, then $P(X < a) = 0.05$ and $P(X > b) = 0.05$, as shown in the figure below.

```
In [9]: from IPython.display import Image
Image('GaussianRV.jpg',width=300)
```

Out [9]:



The distance from the mean to a is the same as the distance from the mean to b . Thus we really only need to find one of these values in order to know both. Leveraging the Q-function table, I will compute the survival function of X at b , that is, $P(X \geq b) = \int_b^{\infty} e^{-t^2/2} dt$. Recall that continuous random variables have 0 probability at any particular value, $P(X=b)=0$, thus, we can write $P(X>b)=P(X \geq b)$.

From the information given, we want to find b such that $P(X > b) = P(X \geq b) = 0.05$ iff $Q(b) = 0.05$ iff $b = Q^{-1}(0.05) \approx 1.65$. By symmetry, $a = -1.65$.

Let's check using `scipy` functions.

```
In [10]: import scipy.stats as stats

# Defining Gaussian Distribution with mean 0 and variance 1
X = stats.norm(loc=0, scale=np.sqrt(1)) #scale is the standard deviation

# Inverse of Survival function at 0.05
```



```
a = X.ppf(0.05)
b = X.isf(0.05)
print("[a,b] = [",a," ",b,"]")

# Check if  $P(-1.65 \leq X \leq 1.65) = P(X > a) - P(X > b)$ 
print("P(X is in [a,b])=",X.sf(a) - X.sf(b))
```

```
[a,b] = [ -1.6448536269514729 ,  1.6448536269514729 ]
P(X is in [a,b])= 0.9000000000000001
```
